From Plant to Patient
Catching launch risk before it costs €7.9M: governed, self-service AI across demand, supply, manufacturing and reliability

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The question isn't: 'do we have the data?'

can a business leader get a trustworthy answer in seconds, in their own language, without waiting on a report?

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Executive summary

What we solved, and what it is worth

Problem: Zyvanta, our most important immunology launch, is at 33% readiness. Master data and artwork blockers only surface at the monthly S&OP, and the Valmora coater that makes it is wearing out.

Outcome: one governed Control Tower. Risk, then why, then the hidden supply risk, then an owned action, in about 60 seconds, in Italian or English.

Value: about €9.7M a year (estimate): €7.85M launch protection, €1.00M OTIF, €731k downtime, €107k expiry.

Ask: a 6-week pilot on the next two launches. Success = every blocker flagged at least 4 weeks before the S&OP review.

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Why launches slip today

The Helivara situation as of 22 September 2026

Launch readiness lives in three places: material master (ERP), artwork approval and serialization. It gets reviewed once a month.

Zyvanta: master data 0%, artwork 0%, serialization 100%, so readiness 33%.

Manufacturing doesn't know which machine a launch depends on: the Valmora coater has a 94% chance of an unplanned failure within 30 days.

"Is my launch safe, and why?" takes days of report chasing across systems, and the answers don't always agree.

Every week of Zyvanta delay costs about €1.96M of revenue (4,361 packs × €450).

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Quantified value

About €9.7M a year: estimates computed from the data

Lever	Annual value (estimate)	How it's computed	KPI owner
Launch protection (Zyvanta)	€7.85M	4,361 packs/week × €450 × 4 weeks earlier	COO / Head of Global Supply Chain
OTIF improvement	€1.00M	5 OTIF points × €200k per point	Supply chain / commercial
Unplanned downtime avoided	€731k	unplanned hours/yr × €8,000/h × 20%	Plant & reliability
Expiry write-off avoided	€107k	units at risk/yr × €12 × 25%	Finance
Total	**€9.7M**	**assumptions editable in table value_assumptions**	

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Value for sponsor and owner

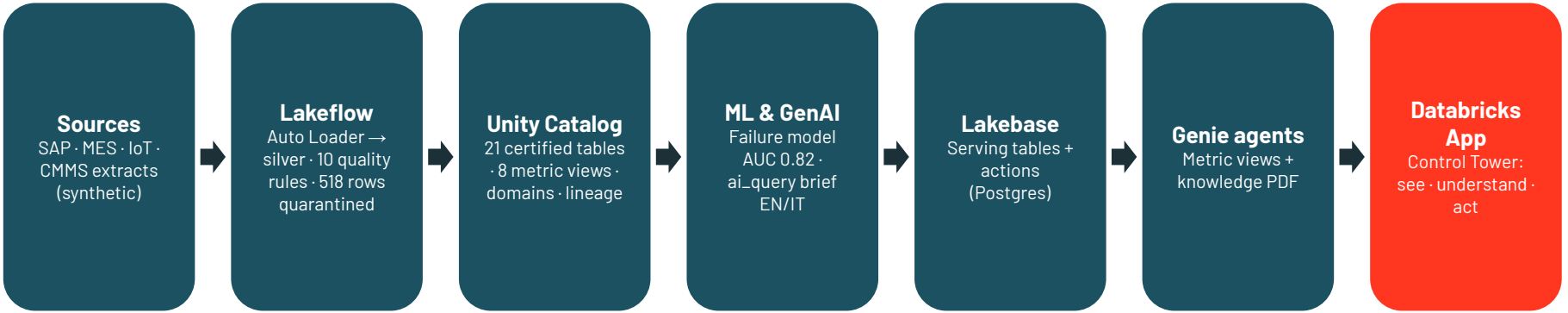
Two altitudes, one platform

Executive sponsor: COO / Head of Global Supply Chain	Domain owner: Global Demand & S&OP Manager
Launch revenue protected (€7.85M on Zyvanta) OTIF +5 pts and fewer tender penalties Lower write-offs and unplanned downtime One governed platform: lower TCO, faster AI	Launch gates every day, not once a month One certified definition of readiness and accuracy Ask in Italian and see the SQL behind the answer Every escalation owned, dated and auditable

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One integrated journey

One governed copy of the data, from raw files to a decision in the app

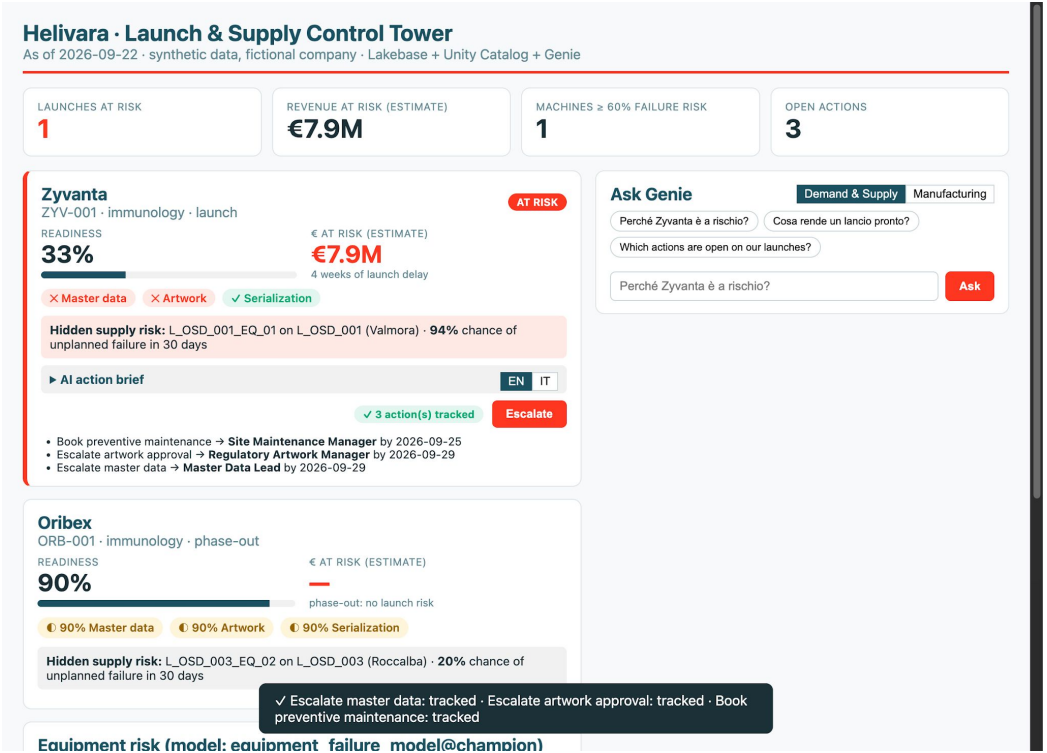


↺ Actions taken in the app are copied back to the lakehouse every day, so Genie and the dashboard see what was decided. One daily Lakeflow Job runs the whole chain: ingest → publish → score → value + brief → Lakebase → copy back.

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Demo!

1 · See: Zyvanta at 33% readiness, €7.9M at risk
2 · Understand: Genie in Italian → master data + artwork; the coater at 94% failure risk
3 · Act: Escalate → owned, dated actions in Lakebase



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What this means for Helivara

From monthly surprise to daily, owned decisions

From monthly surprise to daily signal: blockers visible at least 4 weeks earlier, about €7.85M protected on Zyvanta.

From "which report is right?" to one certified number: Genie answers from governed metric views and shows its SQL.

From insight to accountability: every escalation has an owner, a due date and an audit trail in Lakebase.

From reactive to predictive maintenance: 20% fewer unplanned hours, about €731k a year, and the launch supply protected.

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Decisions and trade-offs

Why this approach and not another

Decision	Why	Trade-off we accepted
Lakeflow feeds the existing governed gold tables	Keeps certification, domains, Genie and metric views intact; quality rules + quarantine	Lineage passes through a publish step
Lakebase for the app	Millisecond reads, transactional write-back, no duplicate escalations	A sync job (managed synced tables need a catalog we couldn't create here)
Classical ML for failure risk, LLM only for the brief	Calibrated, explainable probabilities; LLM grounded on governed facts only	Needs labelled failures; retrain monthly
Genie on metric views, as the signed-in user	Same numbers as finance; per-user permissions	Metric views to maintain as the business changes
Assumptions in a table, not in code	Business can challenge every euro figure	Estimates until the pilot measures them

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Security, governance and cost

Built to pass IT and quality review

Identity: Genie and SQL run as the signed-in user (Unity Catalog permissions per person); the app has least privilege in Lakebase.

Data quality enforced: 10 pipeline rules, 518 bad rows quarantined, none reached the governed tables.

Governance: certified tables and metric views, domains, lineage, and the model versioned in Unity Catalog.

Cost and compliance: serverless and pay per use; Delta history, audit logs and Asset Bundles give a GxP-ready path.

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Next steps: a 30 / 60 / 90-day pilot

Prove the value on two real launches

30 days: connect the material master and artwork systems with Lakeflow Connect, and run the Control Tower on the next two launches.

60 days: stream Valmora line telemetry, retrain the failure model on real work orders, and use the Control Tower in the monthly S&OP.

90 days: measure readiness lead time, OTIF and unplanned hours against the baseline, then decide on scaling to all launches and plants.

The ask: an executive sponsor, the Global Demand & S&OP Manager as domain owner, two launches, and read access to the source systems.

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Appendix
Platform overview and supply-chain glossary

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databricks | DATA+AI PLATFORM | Context. Control. Cost. Choice.

Agentic Apps
Deploy agents at scale to transform work

- **AI/BI** — Business Intelligence
- **Apps** — Custom Apps
- **Lakewatch** — Agentic Security
- **CustomerLake** — Agentic Marketing

Agentic Work
Data-smart coworkers for every employee

Genie
- **ONE** — AI coworker connected to all your data, everywhere
- **AGENTS** — Anyone can build agents to automate work
- **CODE** — Automate building and maintaining data infrastructure
  - ZeroOps
  - App Builder

Agent Bricks
Open agent developer platform

Omnigent — Build | Deploy | Optimize | Govern

Agentic Dev
Develop and deploy agents, apps, and models

Genie Ontology

Unified Governance
Data + AI control and cost management

- **Unity Catalog** — Access | Semantics | Observability | OpenSharing
- **Unity AI Gateway** — Models | Security | Cost Optimization

Agentic Data
Unified, real-time data foundation

- **Lakeflow** — Ingest | Streaming | ETL
- **Lakehouse** — SQL Data Warehousing
- **Lakebase** — Serverless Postgres

Open Infrastructure
Open Format Data Lake — DELTA LAKE | ICEBERG

Any Cloud | Any Model | Any Data

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AI use cases across the life sciences value chain

Increase R&D Productivity

Drug Discovery

- Training Single Cell Models to Predict Gene Expression
- Genetic Target Identification
- Biomedical Research Assistant

Clinical Development

- Clinical Trial Protocol & Result Summarization
- Adverse Event Detection
- Regulatory Document Generation

MERCK, novo nordisk, GILEAD, RGC Regeneron Genetics Center, BAYER

Optimize Supply Chain & Manufacturing

Product Operations

- Digital Twins & Manufacturing Yield Optimization
- Predictive Maintenance
- Demand Forecasting & Safety Stock

VERTEX, Lilly, Johnson&Johnson

Improve Commercial Effectiveness

Sales & Marketing

- Sales Rep Next Best Action to Optimize HCP Engagement
- Medical Affairs Knowledge Assistant
- Marketing Content Generation

AMGEN, GSK

Patient Engagement

- Patient Next Best Action from Digital Devices
- Personalized Patient Adherence Programs
- Patient Education Content Generation

Insulet, welldoc

Optimize Supply Chain & Manufacturing

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From pill to pharmacy: creating competitive advantage through resilient supply chains



94% of life sciences companies agree that the digitalization of the supply chain can make a real contribution to the bottom line; moreover, it can also play a key role in the fulfillment of wider business objectives."

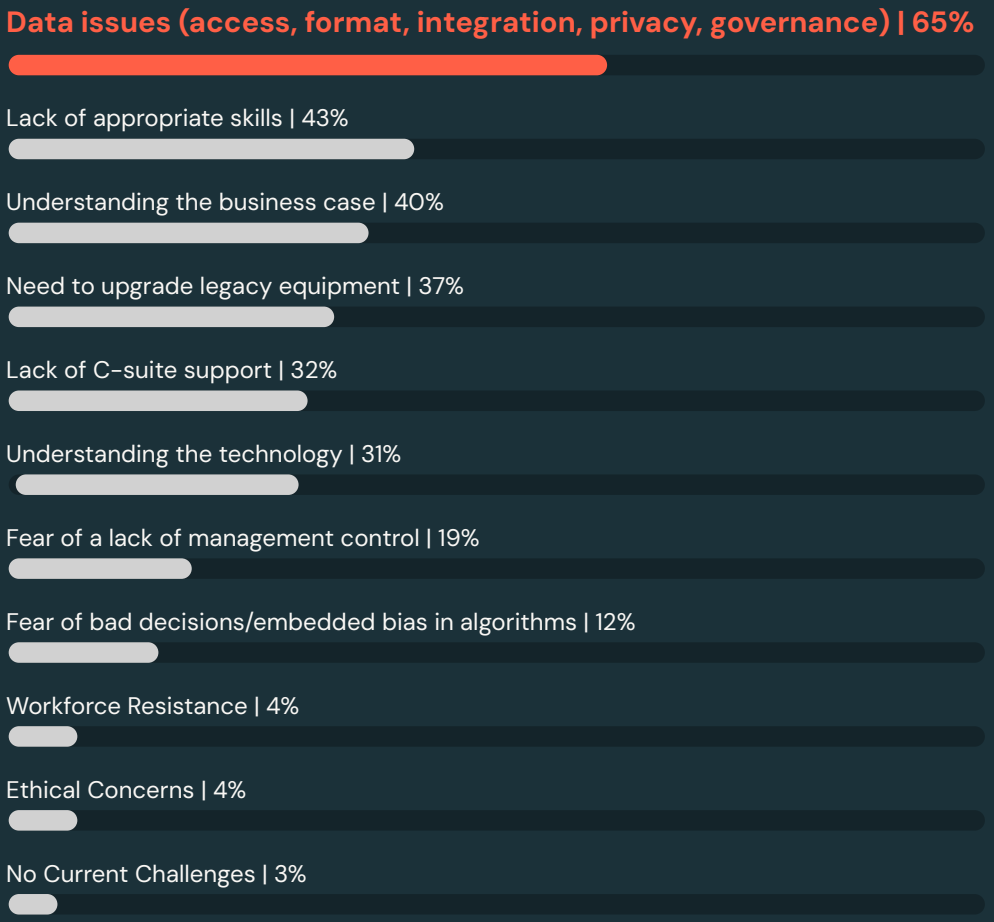
Deloitte.

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Data governance issues dominate challenges of AI Adoption

What do you see as the biggest challenges of AI adoption in your organization today?

MANUFACTURING LEADERSHIP COUNCIL
NATIONAL ASSOCIATION OF MANUFACTURERS



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Current limitations in supply chain tech

Monolithic tools disconnected from real-world cycles



Traditional tools fail to capture real-time market changes & disruptions



Lack sophisticated analytical & AI/ML capabilities



Fragmented data estates that hinder cross-functional collaboration



Proprietary formats and closed ecosystems hindering diverse non-ERP data sources

69%

of chief operations and supply chain officers report their tech investments haven't fully delivered the expected results

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Components of a digital supply chain

Real-Time Visibility



Develop end-to-end supply chain visibility several tiers deep to help control inventory levels, decrease waste, and increase capacity utilization in real time.

Automated Reporting



Inform your C-suite and partners about supply chain issues to ensure enterprise-wide understanding and strategic preparedness.

Prescriptive Action

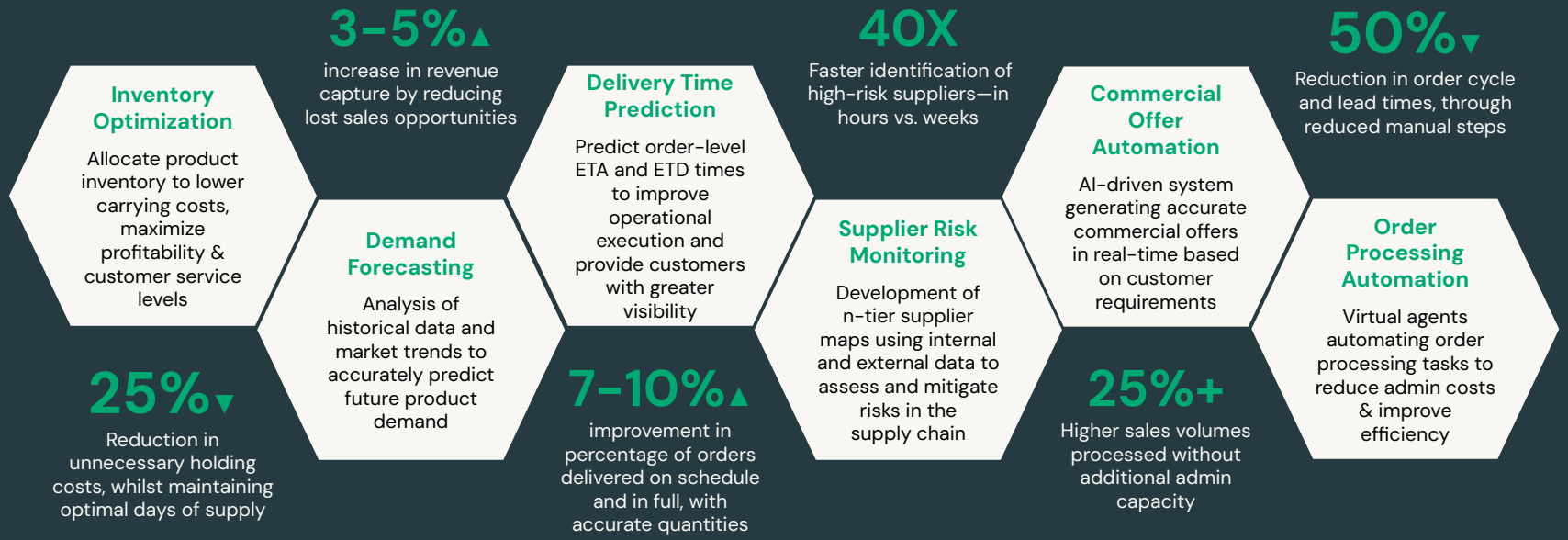


Promote resilience through prescriptive guidance — powered by data and AI — through demand analytics and automated replenishment.

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Key use cases

Validated impact



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Supply Chain

What to consider in the business

Everything it takes to get a product from raw materials to the patient:

1. forecasting demand
2. planning supply
3. making the product
4. storing/moving it
5. delivering it



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Supply Chain - Demand Forecasting

What to consider in the business

Statistical forecast —> the baseline the system generates from history/algorithms.

Consensus forecast —> the final agreed number after humans review and adjust the statistical one (in the S&OP meeting).

Forecast accuracy —> how close the forecast was to actual sales. Often shown as 1 - MAPE (MAPE = Mean Absolute Percentage Error, the average % you were off). Higher accuracy = better.

Forecast bias —> the direction of the error. Positive bias = you consistently over-forecast (predict more than sells → excess stock, expiry). Negative bias = you under-forecast (→ stockouts).



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Supply Chain - Inventory

What to consider in the business

OTIF (On-Time-In-Full) —> the service-level KPI: % of orders delivered on time AND complete.

Stockout —> running out of a product

Backorder —> an unfilled order waiting for stock.

Days of Inventory (DIO / DOH) —> how many days of demand your current stock covers

Shelf-life / expiry-at-risk —> expiry-at-risk = stock likely to expire before it sells (pure waste). The core pharma tension: stockout vs. expiry.

Cold chain —> products that must stay refrigerated end to end.

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Manufacturing

What to consider in the business

OEE (Overall Equipment Effectiveness) —> the master factory KPI = Availability × Performance × Quality.

Availability —> was the line running when it should be? (downtime hurts this)

Performance —> did it run at full speed?

Quality —> how much output was good vs. scrap?

Yield —> actual good output vs. the theoretical maximum from the inputs.

Right-First-Time (RFT) —> % of batches made correctly with no rework/deviation.

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Governance – before launching

What to consider in the business

Launch / renovation —> bringing a new product to market, or updating an existing one (new formulation/pack). "Renovation" = a refresh, not a brand-new product.

Material master data —> the complete, official product record in the ERP (codes, pack size, shelf-life, etc.). If it's incomplete, you literally can't launch.

Serialization —> putting a unique serial number on each pack for track-and-trace (mandated by the EU Falsified Medicines Directive / FMD) to fight counterfeiting.

Artwork —> the printed packaging/labels; must be regulatory-approved before you can produce.

Launch readiness —> a combined score of whether master data + serialization + artwork are all ready. Low score = launch at risk.



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